INTRODUCTION

Background of Case Study

ClariFi is a voice-first personal expense-tracking app with a mobile client and backend API that makes expense tracking fast and turns those records into a monthly expense analysis. Users can log spending via speech or receipt OCR, confirm structured entries, and view the ledger and monthly spending. Users can also contribute supermarket brochures and promotional flyers through a gamified system that rewards consistent uploads with redeemable vouchers, helping expand community-driven price data. On top of tracking, ClariFi helps users decide where and when to buy through price history, comparisons, promotion ingestion, and alert notifications, with roadmap support for bill splitting and family or shared budgeting.

CHOSEN SDG

SDG 10 – Reduced Inequalities

PROBLEM STATEMENT

Sustainable Development Goal 10 (SDG 10) targets the reduction of inequalities within communities through empowerment and promotion of the social, economic and political inclusion of all parts of society, regardless of their racial background or economic status. Among the inequalities, socioeconomic inequality and the digital data gap have caught our group's attention. In this digital age, these targets are increasingly dependent on "fintech" or financial technology, which catalyses financial inclusion by providing scalable, low-cost, and user-friendly solutions to the underserved. We aim to eliminate "data darkness" that prevents households from identifying leakages, planning for the future, or making informed purchasing decisions. A lack of historical insights and high-friction spending-tracking tools is a major hindrance to Malaysians' ability to better understand their spending patterns and cash outflows. This problem is most noticeable among students, young adults, and budget-conscious households who experience frequent daily spending on necessities such as meals, transport, and groceries. A complicated and time-consuming process, which is used in current applications, hinders the group from recording its expenses. Without a convenient way to record transactions in real time, they lose visibility over their spending patterns and struggle to make financial decisions.

Furthermore, they suffer from the price information asymmetry, forcing them to overpay for essential goods (like rice, oil, and eggs) because they lack the data to know where items are cheaper or when prices are trending upward. They are forced to make purchase decisions based on proximity and habit rather than market intelligence. This issue is particularly shown in grocery shopping, where prices for the same item may vary significantly between supermarkets or change frequently due to promotions and weekly flyers. The absence of financial recording applications integrated with transparent price data discourages consumers from making informed choices and optimising their limited resources. This resulted in a decrease in purchasing power and real wages.

OBJECTIVES

1. **REDUCE FRICTION IN EXPENSE TRACKING**

   Cut expense logging from minutes to seconds with voice and receipt-first capture, reducing the barrier of entry for users to record expenses.

2. **TRANSFORM UNSTRUCTURED SPENDING DATA INTO ACTIONABLE INSIGHTS**

   Turn raw spending data into trusted, structured insights users can easily understand and act on immediately.

3. **IMPROVE FINANCIAL DECISION-MAKING THROUGH PRICE TRANSPARENCY**

   Using live data (*localised price comparisons, promotion tracking, and price drop alerts*), help users make informed purchasing decisions and optimise their daily spending.

4. **REDUCING INEQUALITIES AS ALIGNED WITH SDG10**

   Bridging the gap by enabling users to track expenses with low learning cost and easily make informed purchasing decisions.

REVIEW OF EXISTING SOLUTIONS

Analyse current solutions or competitors in the market.

SOLUTIONS	COMPETITORS
Budgeting-first apps	**YNAB:** Very strong for intentional budgeting and behaviour change; weaker on fast capture via voice or OCR and local market pricing **Rocket Money:** Strong on subscription tracking, account aggregation and bill negotiation; weaker on receipt or voice-first capture and item-level local price intelligence
Expense-capture apps	**Expensify:** Best-in-class for receipt capture, reimbursements, approvals, accounting workflows; not community price intelligence
Shared-expense apps	**Splitwise:** Dominant in friend or group bill splitting and balances and has receipt scanning; not full personal finance and price intelligence
Price or deal-discovering apps	**Flipp:** Strong weekly circulars, coupons or local deal discovery; mainly focus on the US and Canada **ShopSavvy:** Strong barcode search, cross-retailer price tracking, and alerts **Basket:** Grocery list and cross-store grocery price comparison with

	a crowdsourced angle

Highlight gaps or limitations that your project aims to fix.

1. “Pre-Purchase” Intelligence Gap
    - Help users decide where to shop by comparing item prices using data collected from brochures, flyers, and receipt OCR.
2. Data Reliability
    - Speech-to-Text (STT) is excluded from the price comparison functionality.
3. Family Profile
    - Family members can share a family profile to track and view each other’s spending and share promotions or deals they discover.

METHODOLOGY

Technical Track

1. ClariFi was developed as a TypeScript monorepo with three main parts:
    - Mobile
    - API

    The mobile client is built with Expo React Native to support a fast voice-first and camera-first experience on one codebase, and the backend is built with NestJS and organised into modular domains such as authentication, parsing, expenses, reports, prices, notifications, families, splits, and subscriptions.

2. Libraries and SaaS usage:

- Authentications: Clerk
- Documents storage(audio, receipts, and document images): Supabase Storage
- Data processing services: cloud speech-to-text, OCR, and LLM-based document parsing.
- Database: PostgreSQL, Prisma

3. The implementation was done in stages. First, we set up the project foundation: authentication, environment validation, database schema, and shared contracts. Next, we implemented the core expense capture flow, including voice capture, receipt upload, OCR/parsing, and user confirmation before saving data. After that, we built the reporting and price-intelligence layer, which includes monthly summaries, price comparison, history, promo ingestion, and alerts. Finally, we added collaboration and productisation features such as family groups, bill splitting, push notifications, and subscription entitlement limits. We also used automated tests and build checks throughout development to ensure the major modules worked together reliably.

4. Data Streamline

```
                +----------------------+
                |      User Input      |
                |----------------------|
                |  Voice Logging       |
                |  Receipt Upload OCR  |
                |  Promo / Flyer Upload|
                +----------+-----------+
                           |
                           v
                +----------------------+
                |   Data Processing    |
                |----------------------|
                | Speech-to-Text       |
                | OCR Parsing          |
                | Data Structuring     |
                +----------+-----------+
                           |
          +----------------+----------------+
          |                                 |
          v                                 v
+-------------------------+      +----------------------------+
| Personal Expense DB     |      | Verified Price Database    |
|-------------------------|      |----------------------------|
| Voice expense records   |      | Receipt OCR prices         |
| User spending history   |      | Flyer / promo data         |
| Monthly analysis        |      | Community submissions      |
+-----------+-------------+      +-------------+--------------+
            |                                  |
            v                                  v
+-------------------------+      +----------------------------+
| Expense Insights        |      | Price Intelligence Engine  |
|-------------------------|      |----------------------------|
| Spending reports        |      | Price comparison           |
| Budget tracking         |      | Promotion alerts           |
| Financial visibility    |      | Price trend analysis       |
+-------------------------+      +----------------------------+
```

Non-Technical Track

1. Design Thinking Process : User-centered Process
    - Focus on busy users who want to track spending with minimal effort.
    - Key pain points were manual entry friction, lack of actionable insights, and no easy way to manage shared expenses.
    - We prioritized fast capture (voice and receipt), clear visibility (monthly expense analysis), and decision support (price alerts and signals).
    - We shaped the app around real usage moments, not around technical modules.
    - We refined flows to reduce taps, simplify wording, and keep users moving forward with clear feedback.

2. User Research Approach
    - ➢ Identified user segments
        - Individuals tracking personal spending
        - Families or friends splitting expenses
        - Budget-conscious users comparing prices and promotions
    - ➢ Gathered core needs
        - “Capture now, classify later” behaviour
        - Confidence that backend sync and health are working
        - Lightweight collaboration for household finance
    - ➢ Turned findings into product priorities
        - Voice or receipt-first capture over long forms
        - Dashboard or reporting that explains “where money went”
        - Price comparison and alerts for savings decisions

3. Flow Creation

➢ Capture flow
  - User records voice or uploads receipt
  - App parses expense and lets user confirm quickly
  - Entry appears in ledger or reporting

➢ Insight flow
  - User opens monthly report
  - Sees category breakdown, top merchants or items, and anomalies
  - Uses insights to adjust spending behaviour

➢ Price decision flow
  - User searches item and area
  - App shows compare and history
  - User creates alerts and receives notification when conditions match

➢ Shared finance flow
  - User creates or joins family
  - Creates split session, assigns participants or allocations, and finalizes balances
  - Group gets clear settlement views

1.2.

RESULTS (SOLUTION)

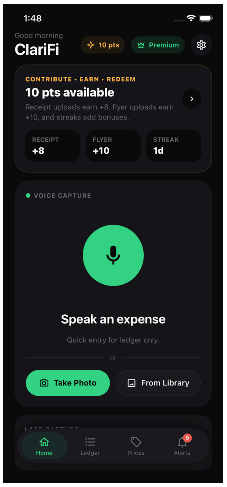
Summarise the outcome of your work. Did you meet your objectives?

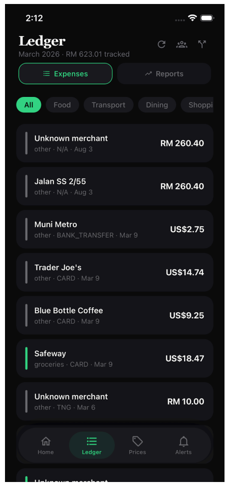
We delivered the core ClariFi product scope, like quick expense capture, monthly expense summary, price alerts, price comparison, and shared family expense workflows. We met the main objective for product design and feature coverage. Sticking close to our objective to create frictionless expense tracking and enabling users to make informed purchasing decisions in order to achieve SDG10, we have implemented the following features.

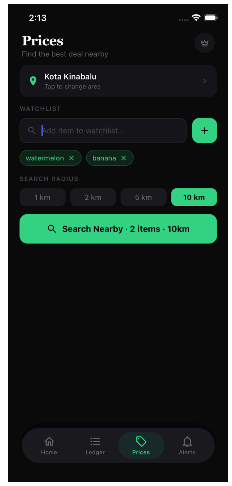
- Voice and receipt parsing implemented
- Ledger and monthly expense summary are functional
- Price collection and history infrastructure is in place
- Price comparison feature is now available for use
- Family wallets and bill splitting are scaffolded and working
- Clerk-based auth with role-based access control is live

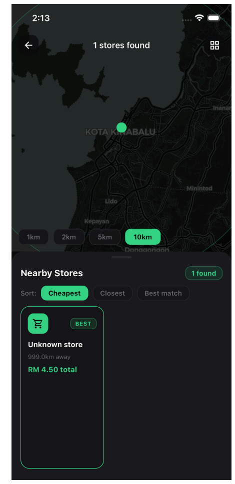
PROTOTYPE SHOWCASE

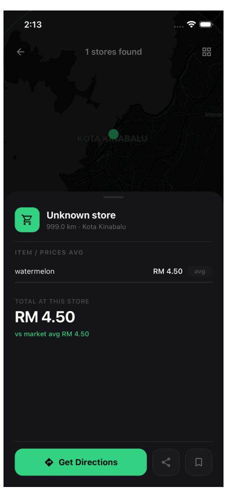
Output / UI Screenshots

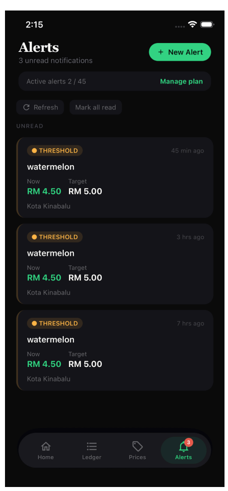












AI Acknowledgement

1. What AI we used
   - OpenRouter-based AI models, including OpenAI GPT-4o-mini-transcribe for speech-to-text and OpenAI GPT-4.1-mini for OCR and document understanding.
   - On-device speech recognition on mobile for fast local voice capture.
   - Claude code for development and testing.

2. Usage
   - Speech-to-text was used to convert spoken expenses into transcript text before structuring them into expense entries.
   - OCR and vision-capable AI were used to extract visible text and item details from receipts and promotional flyers.
   - LLM-based parsing was used to classify uploaded documents as receipts, flyers, or unknown documents, then convert unstructured content into structured fields such as merchant name, amount, date, line items, and promotion details.
   - AI outputs were not stored blindly. Users still reviewed and confirmed parsed results before final submission, which kept a human-in-the-loop validation step in the system.

CONCLUSION

Summarize your project and suggest potential future improvements.

We built ClariFi to make personal and family expense tracking easier in real life, not just in theory. Users can capture expenses quickly through voice or receipts, review monthly spending trends, monitor price changes, do price comparison and manage shared expenses in one place.

Future improvements:

1. Integrations
   - Add bank sync, e-wallet import, calendar and location context to reduce manual input and improve expense accuracy.

2. Better personalization
   - Use user behaviour patterns to improve auto-categorization and set smarter and adaptive alert thresholds.

3. Growth features
   - Introduce gamified savings goals, challenge modes, and household benchmark comparisons to increase engagement and long-term retention.